TCAD Based Study of 4H-SiC Charge Sheet Superjunctions with a Wide-Trench Termination

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Abstract

In a Charge Sheet Superjunction (CSSJ), the p-pillars of a conventional superjunction are replaced by a fixed negative charge at the Al_2O_3/SiO_2 interface of a trench dielectric, which avoids the deep p-pillar formation required in conventional 4H-SiC superjunctions. Work on the CSSJ so far has been limited to the active cell. Here we propose an edge termination for 4H-SiC CSSJs in which a single wide trench receives the same SiO_2/Al_2O_3 stack as the active trenches. The same interface charge is therefore available in the termination, without an additional dedicated termination implantation step. Two-dimensional Sentaurus TCAD simulations were carried out for a 1700 V-class device whose active cell breaks down at about 1.9 kV. Without a field plate, the termination saturates at about 940 V because of field crowding at the trench corner. A field plate over the trench, placed on a thin dielectric, splits the peak field between the trench corner and the plate edge. With a trench width of 18 µm and a field plate length of 6.5 µm, the termination reaches about 95 % of the nominal active-cell breakdown voltage. Breakdown voltage rises with temperature at about 0.24 V/K, consistent with avalanche-dominated breakdown. A compatible fabrication sequence using the same trench and dielectric-stack formation is also outlined.

1. Introduction

Power converters in electric vehicles, solar inverters and motor drives rely on semiconductor switches that block high voltage with little leakage when off and conduct with a low specific on-resistance, $R_{ON,sp}$, when on. In a unipolar device the same drift layer has to do both jobs, and this ties $R_{ON,sp}$ to the breakdown voltage through the unipolar limit [1]

$$R_{ON,sp} = \frac{4\,V_{BR}^2}{\mu_n\,\varepsilon_s\,E_c^3}, \tag{1}$$

where V_{BR} is the breakdown voltage, μ_n the electron mobility, ε_s the permittivity and E_c the critical field. 4H-SiC has a critical field of 2.5–3 MV/cm, close to ten times that of silicon, together with a 3.26 eV bandgap and roughly three times the thermal conductivity. Its $R_{ON,sp}$ at a given V_{BR} can be more than two orders of magnitude lower [1], but Eq. (1) still scales with V_{BR}^2.

A superjunction (SJ) gets around this limit with charge-balanced n- and p-pillars [2]. In 4H-SiC, however, deep p-pillars can only be made by repeated epitaxy, sidewall implantation or epitaxial trench refill [3, 4]. Our group earlier proposed an alternative drift-layer architecture, the Charge Sheet Superjunction (CSSJ), in which the p-pillar is replaced by a bilayer Al_2O_3/SiO_2 insulator carrying a fixed negative charge at the interface between the two layers [5, 6]. The group has since studied the CSSJ in 4H-SiC in detail, covering its practicability, analytical modelling and design [7, 8, 9], and a 1.2 kV CSSJ trench MOSFET with its process design [10]. These studies show that in 4H-SiC the CSSJ needs no p-pillar, can reach 5–45 % lower $R_{ON,sp}$ than an equivalent SJ, and tolerates charge imbalance more than ten times better than its silicon version. The concept of using a dielectric charge to mimic the p-pillars has recently been developed commercially in silicon and is the foundation of the SuperQ™ technology from iDEAL Semiconductor [11].

Any real device also needs an edge termination, or it breaks down early at its periphery (Fig. 1). *Floating Field Rings* (FFR/FLR) drop the voltage in steps across a set of floating p-rings. They tolerate process variation well but take up a lot of area [12]. *Junction Termination Extension* (JTE) uses one lightly doped p-region and a single implant, but its performance depends strongly on the dose [13]. *Bevel* terminations need an angled edge and special dicing, while *field plates* extend the electrode over a dielectric to move the equipotential lines away from the junction. Planar schemes of this kind usually need several times the drift thickness, which is a poor fit for the heavily doped drift layer of an SJ. In *trench terminations* a deep dielectric-

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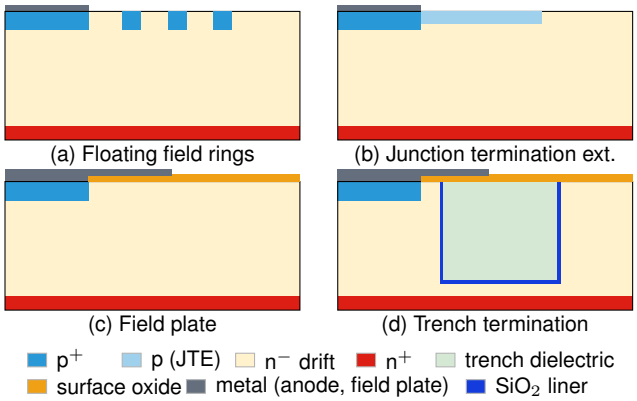


Fig. 1 Common edge termination techniques for power devices.

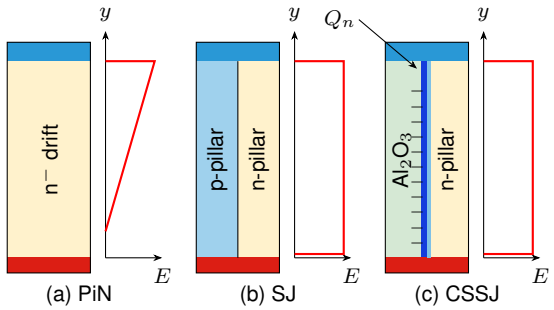


Fig. 2 Half-cells and vertical field profiles at breakdown: (a) conventional drift layer with triangular field; (b) superjunction; (c) charge sheet superjunction, where the negative sheet charge (SiO_2 liner shown in blue) induces a hole layer that replaces the p-pillar.

filled trench takes up most of the voltage, and the termination becomes much narrower [14, 15].

All of these CSSJ studies, however, have focused on the active region. To the best of our literature survey, an edge termination specifically designed for a 4H-SiC CSSJ has not yet been reported. In this paper, we propose a wide-trench termination that uses the dielectric stack and interface charge already present in the active cell, and we evaluate it using TCAD.

2. Device Operation

2.1 Superjunction

In a conventional drift layer the field falls linearly away from the p^+n junction (Fig. 2a), and much of the layer carries little field. In an SJ, alternating n- and p-pillars of widths W_n, W_p and doping N_d, N_a satisfy the charge-balance condition $N_d W_n = N_a W_p$. Under reverse bias the pillars deplete each other sideways at a low voltage. After that, the net charge in the drift region is close to zero and the vertical field is nearly flat (Fig. 2b). V_{BR} then approaches $E_c \cdot D_{Drift}$ and no longer depends on the pillar doping, so the n-pillar can be doped much more heavily. For a cell width W, the on-resistance scales as [2]

$$R_{ON,sp} \propto \frac{W\, V_{BR}}{\mu_n\, \varepsilon_s\, E_c^2}, \tag{2}$$

which grows linearly rather than quadratically with V_{BR}. However, this approach reduces the active conduction area and introduces fabrication challenges associated with p-pillar formation in SiC, while remaining sensitive to charge imbalance [16].

2.2 Charge Sheet Superjunction

In a CSSJ (Fig. 2c), the p-pillar is replaced by a thin SiO_2 liner and an Al_2O_3 layer. The Al_2O_3/SiO_2 interface carries a fixed negative charge per unit area, Q_n, whose magnitude can be set between about 2.5 and $7.9 \times 10^{12}\,\mathrm{cm}^{-2}$ by the Al_2O_3 deposition temperature [17, 8]. From Gauss's law, the charge sets up a field of about qQ_n/ε_{SiC} normal to the sidewall. This field inverts the n-pillar surface, and the hole layer connects to the p^+ anode [8, 10]. Under reverse bias the hole layer depletes the n-pillar from the side in the same way a p-pillar does, but it takes up no silicon carbide area. Charge balance requires

$$Q_n = N_d \cdot W_n, \tag{3}$$

where W_n is the width of the n-region compensated by each charge sheet. Any mismatch is described by the imbalance factor [8]

$$k_{eff} = 1 - \frac{Q_n}{N_d W_n}, \qquad Q_n \le N_d W_n. \tag{4}$$

The sensitivity of V_{BR} to k_{eff} scales as $1/E_c$, so in 4H-SiC it is over ten times smaller than in silicon [8].

In an SJ the p-pillar carries no current, whereas in a CSSJ it is replaced by an insulator of width W_I that can be made much thinner than a p-pillar. With W_{d0} the zero-bias depletion width of the induced junction and L the drift length, the on-resistance is [8]

$$R_{ON,sp} = \frac{L\,(1 + W_I/W_n)}{q N_d \mu_n \left(1 - \sqrt{2}\, W_{d0}/W_n\right)}, \tag{5}$$

and for $W_n \gg \sqrt{2} W_{d0}$ the ratio to an SJ of the same doping becomes

$$\frac{R_{ON,sp,CSSJ}}{R_{ON,sp,SJ}} \approx 0.5 \left(1 + \frac{W_I}{W_n}\right). \tag{6}$$

For the cell used here ($W_I/W_n \approx 0.37/1.57$), Eq. (6) gives about 0.62, or close to 40 % less resistance than the SJ. The CSSJ keeps the flat SJ field when off and gives almost the whole cell to the current when on.

2.3 Importance of Edge Termination

The flat field described above exists only inside the active area. At the edge, the last pillar has no balanced neighbour, so the equipotential lines bend up towards the surface and bunch together at the last junction or trench corner. Avalanche then starts well below the ideal voltage. A termination has to spread these lines out over as short a distance as possible, and its performance is usually given as the termination efficiency. For the present study, this efficiency is evaluated relative to the nominal active-cell reference voltage,

$$\eta = \frac{V_{BR,term}}{V_{BR,ref}} \times 100\,\%. \tag{7}$$

and by its width, which in SiC directly adds to the costly die area: a 14 µm trench termination raised the area efficiency of a 2 A SiC diode from 62 % to about 90 % compared with FLR/JTE [18].

2.4 Previous Work on Trench Termination

A trench termination moves part of the voltage out of the semiconductor and into a dielectric, which can hold a much higher field without avalanche. The Deep Trench Termination (DT2) filled a large, deep trench with benzocyclobutene and added a field plate, reaching nearly 1300 V in silicon [14]. Universal design curves from more than 800 simulated configurations showed that deeper trenches allow shorter terminations [19]. Deep trenches were then applied to 600–1200 V silicon SJ diodes [16, 20]. Trench field plates also shape the off-state field in silicon MOSFETs [21] and gave 1.66 kV in β-Ga_2O_3 diodes [22].

In 4H-SiC the high E_c puts heavy stress on the trench dielectric. Wang et al. encapsulated a spin-on dielectric with SiO_2 and obtained 1750 V, 96 % of the ideal value, with a 14 µm termination, less than a quarter of an FLR [15]; they then showed that the field is terminated by holes accumulated at the outer trench sidewall [18]. For SiC SJ devices, a wide-trench termination raised the simulated V_{BR} of an SJ Schottky diode from 1423 to 2600 V for trench widths above 20 µm, and a narrower outermost mesa reduced field crowding at its top [3]. A ring-assisted SJ termination extension reached 4.15 kV in an epi-refilled SJ diode [4], and a high-k TiO_2 insert in an SiO_2-filled trench moved the field peak away from the active-region corner to improve avalanche ruggedness [23]. None of these was designed for the CSSJ, where the active trenches already contain a charged dielectric.

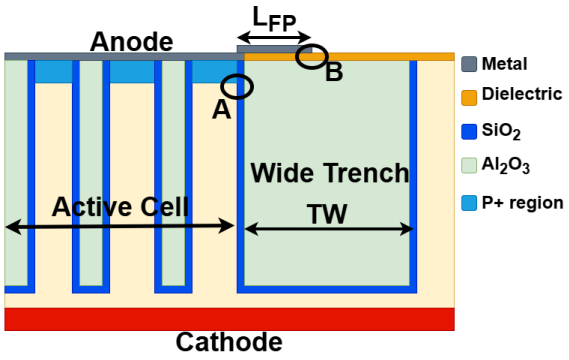


Fig. 3 Schematic cross-section of the off-state equivalent of a CSSJ diode/MOSFET with the proposed wide trench termination. A and B mark the locations of the electric field peaks.

2.5 Proposed Termination

Figure 3 shows the structure we propose. In the active cell, n-mesas sit between narrow trenches lined with SiO_2 and filled with Al_2O_3. The termination is a single wide trench of width TW next to the last active mesa, with the same depth $D_T = 15\,\mu\text{m}$ and the same bilayer. The drift thickness is $D_{Drift} = D_T + k$ with $k = 1\,\mu\text{m}$. A thin dielectric covers the trench, and the anode metal extends over it by L_{FP} to form a field plate. The wide trench gives the equipotential lines room to spread sideways through the dielectric. The charge on the trench sidewall depletes the outer edge of the last mesa, analogous to the charge-compensation effect in the active cell. This reduces the required termination length. The field plate then keeps the lines away from the top corner of the trench. Because the normal displacement is continuous at the SiO_2/SiC interface ($\varepsilon_{SiC} E_{SiC,\perp} = \varepsilon_{ox} E_{ox,\perp}$), the normal field in SiO_2 is about 2.5 times that in SiC, and the dielectric takes a large share of the voltage.

3. Simulation and Results

3.1 Simulation Setup

We ran all simulations in two dimensions in Synopsys Sentaurus TCAD (version 2023.12) [24]. Impact ionisation uses the anisotropic Hatakeyama model for 4H-SiC [25]; the Okuto–Crowell model gave consistent results. Doping- and field-dependent mobility, SRH and Auger recombination are included. The fixed charge is placed at the Al_2O_3/SiO_2 boundary of every trench, with a thin SiO_2 interlayer between SiC and Al_2O_3 (Fig. 4). Perfect balance corresponds to $7.84 \times 10^{12}\,\text{cm}^{-2}$;

Table 1 Device and simulation parameters.

Parameter	Value
Trench depth, D_T	15 µm
Trench-to-substrate spacing, k	1 µm
Drift thickness, D_{Drift}	16 µm
Drift doping, N_d	$\approx 5 \times 10^{16}$ cm^{-3}
Half-cell n-region width, W_n	1.57 µm
Half-cell dielectric width	0.37 µm
Charge for balance ($k_{eff} = 0$)	7.84×10^{12} cm^{-2}
Nominal Q_n ($k_{eff} = 0.2$)	6.27×10^{12} cm^{-2}
Termination trench width, TW	1–40 µm
Field plate length, L_{FP}	4–7 µm
Dielectric under field plate	10 nm
Temperature	300–400 K

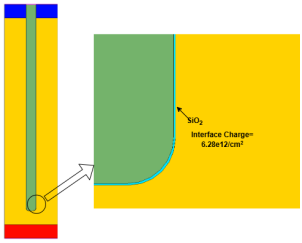


Fig. 4 CSSJ half-cell simulated in Sentaurus TCAD, with a zoomed view of the trench bottom, showing the thin SiO_2 interlayer and the fixed interface charge.

the nominal $Q_n = 6.27 \times 10^{12}$ cm^{-2} ($k_{eff} = 0.2$, a 20 % imbalance) leaves margin for process variation (Table 1).

3.2 Active CSSJ Cell

We first simulated the active cell alone, since it sets the reference for the termination. Its reverse I–V curve (Fig. 5a) breaks down sharply at about 1.87 kV, and the half-cell and full-cell results agree at about 1.9 kV. At breakdown the field along the mesa is nearly flat, as an SJ should be (Fig. 5b). It is about 3 MV/cm near the top of the trench and peaks at about 4 MV/cm where the dielectric meets the n-region at the rounded trench bottom (Fig. 5c). For comparison with the termination results, we use the nominal active-cell breakdown voltage of approximately 1.9 kV as the reference $V_{BR,ref}$.

3.3 Termination without Field Plate

Figure 6 gives V_{BR} of the termination without a field plate. With a 1 µm trench it is only about 300 V: the potential cannot spread laterally beyond the last mesa, resulting in strong electric-field crowding at the upper trench corner (region A in Fig. 3). Widening the trench lets the equipotential lines move into the dielectric, and V_{BR} climbs until the lines have fully relaxed inside the trench,

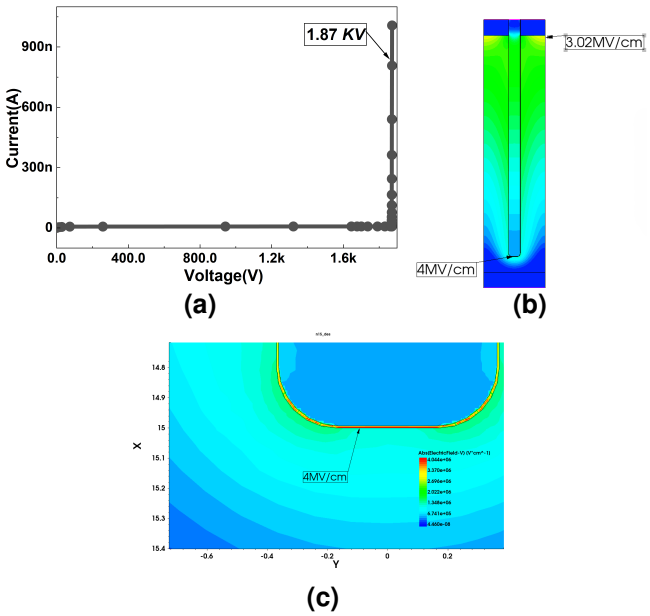


Fig. 5 Active CSSJ cell without termination, simulated in Sentaurus TCAD: (a) reverse I–V characteristic; (b) electric field distribution at breakdown; (c) zoomed view of the trench bottom, where the field peaks at about 4 MV/cm along the dielectric interface.

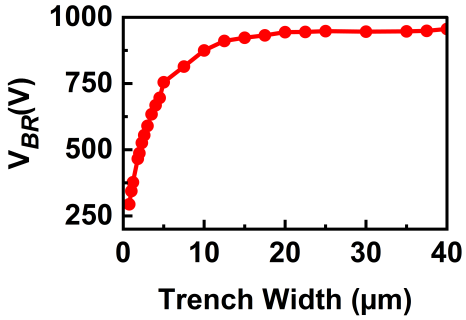


Fig. 6 V_{BR} versus trench width without field plate ($L_{FP} = 0$), simulated in Sentaurus TCAD.

at about $TW = 20$ µm. The plateau is only about 940 V, approximately half of the nominal active-cell reference. Beyond this point the field at A is set by the corner shape, not by TW, so it has to be reduced directly, as also seen in silicon deep trench terminations [14].

3.4 Effect of the Field Plate

Adding a field plate reduces the field at A, but a second peak appears at the plate edge (region B). A thin 10 nm dielectric is placed under the plate as protection against dielectric breakdown.

Figure 7 plots V_{BR} against TW for L_{FP} from 4 to 7 µm. A longer plate lowers the peak at A and pushes the peak at B outwards; the maximum breakdown voltage is obtained when the electric-field peaks at locations A and B become compara-

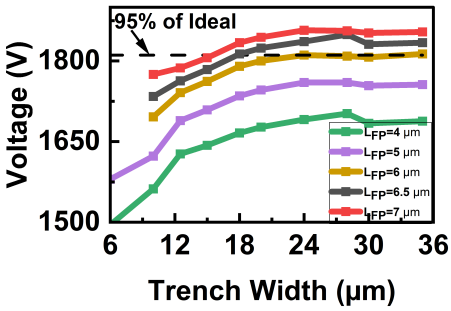


Fig. 7 V_{BR} versus trench width simulated in Sentaurus TCAD for field plate lengths from 4 to 7 µm; the dashed line marks 95 % of the nominal active-cell reference $V_{BR,ref}$.

Table 2 V_{BR} and termination efficiency at $TW = 18\,\mu\text{m}$ (values read from Fig. 7; $V_{BR,ref} \approx 1.9\,\text{kV}$).

L_{FP} (µm)	V_{BR} (V)	η (%)
4	≈ 1665	≈ 87
5	≈ 1735	≈ 91
6	≈ 1790	≈ 94
6.5	≈ 1810	≈ 95
7	≈ 1835	≈ 96

ble. Each curve flattens beyond TW of 18–24 µm for the same reason as before.

Table 2 lists the results at $TW = 18\,\mu\text{m}$. Going from 4 to 5 µm adds about 70 V, but past 6 µm each extra half micrometre gives only 20–25 V. By then the peak at A is already low, and a longer plate only raises the field in the dielectric under its edge and makes the termination wider. We settled on $TW = 18\,\mu\text{m}$ and $L_{FP} = 6.5\,\mu\text{m}$, which gives about 1810 V (approximately 95 % of the nominal active-cell reference) and lies near the knee of both curves.

In its potential plot (Fig. 8) the lines open out smoothly into the trench without bunching at the corner, and the 18 µm trench is comparable to the 16 µm drift thickness.

3.5 Interface Charge and Temperature

We also varied Q_n from 5.45 to $7.84 \times 10^{12}\,\text{cm}^{-2}$, which corresponds to k_{eff} from about 0.3 down to 0 (Fig. 9a). V_{BR} rises almost linearly from about 1480 V to 2630 V, roughly 480 V per $10^{12}\,\text{cm}^{-2}$, and is highest at balance because the increased charge more effectively depletes the mesas and flattens the field. The 1.9 kV reference used for the termination-efficiency calculation corresponds specifically to the nominal $Q_n = 6.27 \times 10^{12}\,\text{cm}^{-2}$ design point; the charge sweep explores other operating points and does not redefine that reference. The termination trench carries the same charge, so it follows the active cell without being re-optimised.

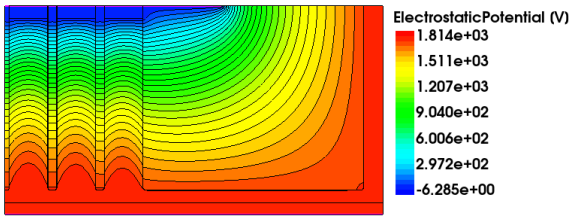


Fig. 8 Potential distribution at breakdown simulated in Sentaurus TCAD for the optimised design ($TW = 18\,\mu\text{m}$, $L_{FP} = 6.5\,\mu\text{m}$).

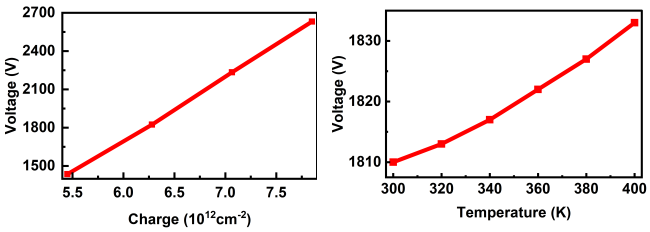


Fig. 9 V_{BR} of the optimised design simulated in Sentaurus TCAD versus (a) negative interface charge and (b) ambient temperature.

Between 300 and 400 K, V_{BR} goes from about 1810 V to 1834 V (Fig. 9b), a coefficient of about 0.24 V/K. A positive coefficient is what avalanche breakdown should show, since stronger phonon scattering at high temperature lowers the ionisation rate [26]. The positive temperature coefficient of breakdown voltage is consistent with avalanche-dominated breakdown within the simulated temperature range [26].

3.6 Comparison with Reported Terminations

In Table 3, the proposed termination has an efficiency and width comparable to a reported SiC p-i-n trench termination [15] and is substantially narrower than the wide trench reported for SiC SJ diodes [3]. It does not require a dedicated p-type termination implant or a separate charge-generating dielectric stack.

4. Possible Fabrication Steps

Because this work is based on TCAD, Fig. 10 shows a proposed fabrication sequence rather than an experimentally demonstrated process. The sequence follows the reported CSSJ process [8, 10] and trench planarisation approach [22]:

1. **Epitaxy:** Grow the n-type drift layer on an n^+ 4H-SiC substrate.
2. **p^+ formation:** Form the p^+ anode by Al im-

Table 3 Comparison of the proposed termination with trench and SJ terminations reported in the literature.

Ref.	Device / termination	Material	V_{BR}	η	Termination width	Type
[14]	p-i-n, BCB-filled deep trench + FP	Si	≈ 1300 V	–	–	Exp.
[22]	SBD, SiO_2-filled trench + staircase FP	β-Ga_2O_3	1660 V	–	–	Exp.
[15]	p-i-n, SiO_2-encapsulated SOD trench	4H-SiC	1750 V	96 %	14 µm	Exp.
[3]	SJ-SBD, wide trench	4H-SiC	1920 V	–	50 µm	Exp.
[4]	SJ-SBD, ring-assisted SJTE	4H-SiC	4150 V	92 %	–	Exp.
This work	CSSJ, charge-sheet wide trench + FP	4H-SiC	≈ 1810 V	≈ 95 %	18 µm	Sim.

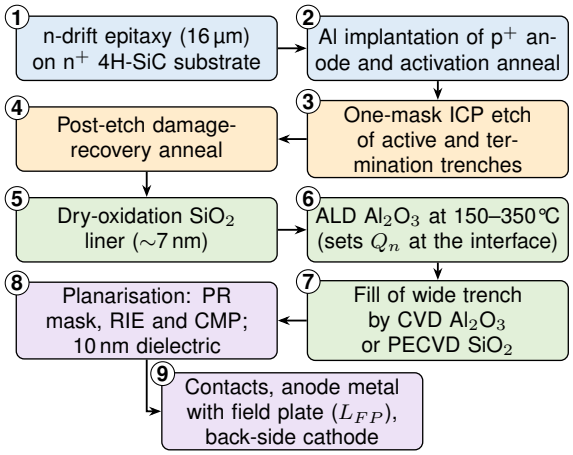


Fig. 10 Possible fabrication flow for the CSSJ with wide-trench termination. The active and termination trenches are formed together, while the termination geometry is defined by the trench and metal layout.

plantation followed by activation annealing.

3. **Trench formation:** Etch the active and termination trenches together using ICP etching and a common trench mask.

4. **Dielectric stack:** Grow the SiO_2 liner and deposit Al_2O_3 by ALD to establish the required interface charge [17, 8].

5. **Trench filling and planarisation:** Fill the wide trench using CVD Al_2O_3 or PECVD SiO_2, followed by RIE/CMP planarisation [22].

6. **Field plate and contacts:** Form the thin field-plate dielectric, pattern the anode metal with overlap L_{FP}, and add the backside cathode.

The termination therefore uses the same charge-generating dielectric stack as the active cell and does not require a dedicated p-type termination implant. The most critical process controls are the Al_2O_3 deposition temperature, which determines Q_n, the trench profile, and void-free filling of the wide trench (Table 4).

Table 4 Key process steps and the parameters that must be controlled.

Step	Main control
Epitaxy	Drift thickness and doping (N_d)
p^+ formation	Implant dose and activation
Trench etch	Depth, sidewall profile, etch damage
Dielectric stack	SiO_2 quality, Al_2O_3 coverage, Q_n (deposition temperature)
Trench fill	Void-free fill of the wide trench
Field plate	Length L_{FP}, dielectric thickness, alignment
Contacts, passivation	Low leakage, high-field stability

5. Conclusion

We have proposed a wide trench edge termination for 1700 V-class 4H-SiC Charge Sheet Superjunctions and studied it by TCAD. The termination trench uses the same SiO_2/Al_2O_3 stack and interface charge as the active trenches, so no additional dedicated charge-generating or p-type termination process is required; its dimensions are set by layout. Without a field plate the termination levels off at about 940 V, limited by the field at the trench corner. A field plate on a 10 nm dielectric splits the peak field between the trench corner and the plate edge, and with $TW = 18\,\mu m$ and $L_{FP} = 6.5\,\mu m$ the termination reaches about 1810 V, approximately 95 % of the nominal active-cell reference, in a width close to the drift thickness. Because the termination and active cell share one charge, they move together when the charge varies, and the positive temperature coefficient of approximately 0.24 V/K is consistent with avalanche-dominated breakdown. Our next steps are to examine the dielectric field under the field plate edge and to verify the design experimentally.

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